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Deep Learning

Project 4

Title: **ETHICAL CONSIDERATIONS IN DEEP LEARNING**

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PROJECT 4

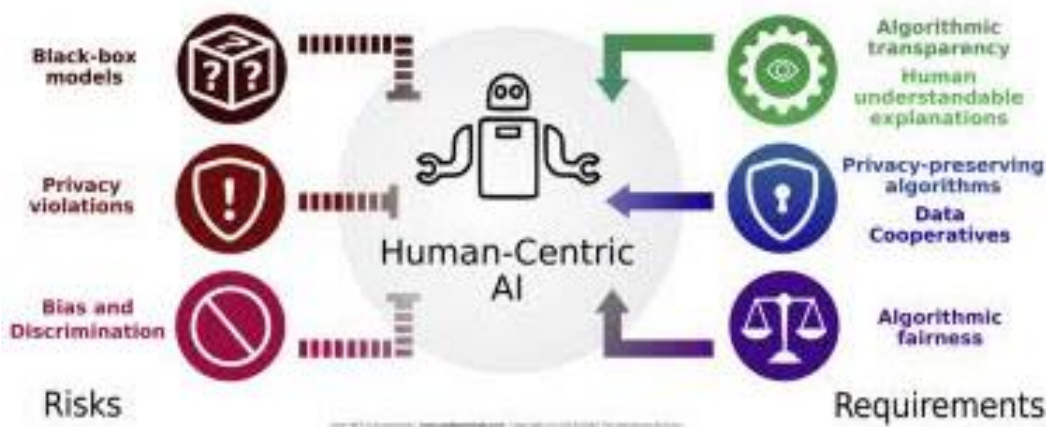
ETHICAL CONSIDERATIONS IN DEEP LEARNING

Exploring bias, privacy, fairness, transparency, and responsible use of AI in Deep Learning systems.

INTRODUCTION

Deep Learning has transformed fields such as healthcare, finance, criminal justice, and hiring by enabling systems to learn complex patterns from large datasets. However, as these systems increasingly influence real human lives, they raise serious ethical questions.

This project examines three major ethical issues in Deep Learning, analyzes a real-world case where an AI system caused ethical concern, and proposes practical measures to address these problems.



TASK 1: THREE ETHICAL ISSUES IN DEEP LEARNING

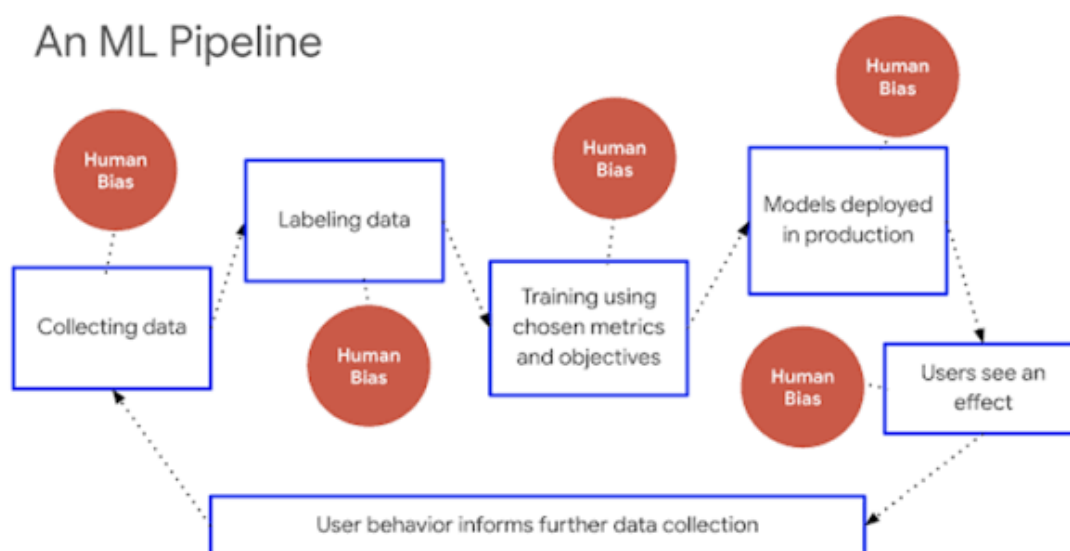
1. BIAS AND DISCRIMINATION

Deep learning models learn statistical patterns from their training data. If that data reflects historical or social biases, the model will reproduce — and sometimes amplify — those biases in its predictions.

This can result in unfair treatment of people based on race, gender, age, or other protected characteristics, even when no one intended discrimination.

Examples

- Facial recognition systems have shown higher error rates when identifying women and people with darker skin tones compared to lighter-skinned men, because training datasets were not diverse enough.
- Hiring algorithms trained on past resumes have been found to downgrade candidates from certain demographic groups because historical hiring data favored a group.



2. PRIVACY VIOLATIONS

Deep learning systems typically require massive amounts of data for training, often collected by scraping the internet, social media, or user records — sometimes without explicit user consent.

This raises concerns about surveillance, data ownership, and the possibility that models may memorize and later leak sensitive personal information.

Examples

- Large image-recognition models have been trained using photos scraped from social media without users' knowledge or consent.

- Language models have occasionally been shown to reproduce memorized snippets of personal data, such as email addresses, that appeared in their training data.

3. LACK OF TRANSPARENCY – THE "BLACK BOX" PROBLEM

Deep neural networks, especially large and complex ones, are difficult to interpret — even their own designers often cannot fully explain why a specific decision was made.

This lack of explainability is especially problematic in high-stakes domains such as healthcare diagnosis, loan approval, or criminal sentencing, where accountability and the ability to challenge a decision matter greatly.

Examples

- A deep learning model used to approve or reject loan applications may reject an applicant without providing a clear, understandable reason, making it hard to detect errors or unfair treatment.
- Medical diagnosis models can flag a scan as "high risk" without indicating which visual features led to that conclusion, making it harder for doctors to trust or verify the result.

TASK 2: REAL-WORLD EXAMPLE – THE COMPAS RECIDIVISM ALGORITHM

COMPAS (Correctional Offender Management Profiling for Alternative Sanctions)

COMPAS is a risk-assessment tool that has been used by courts in parts of the United States to predict how likely a defendant is to reoffend.

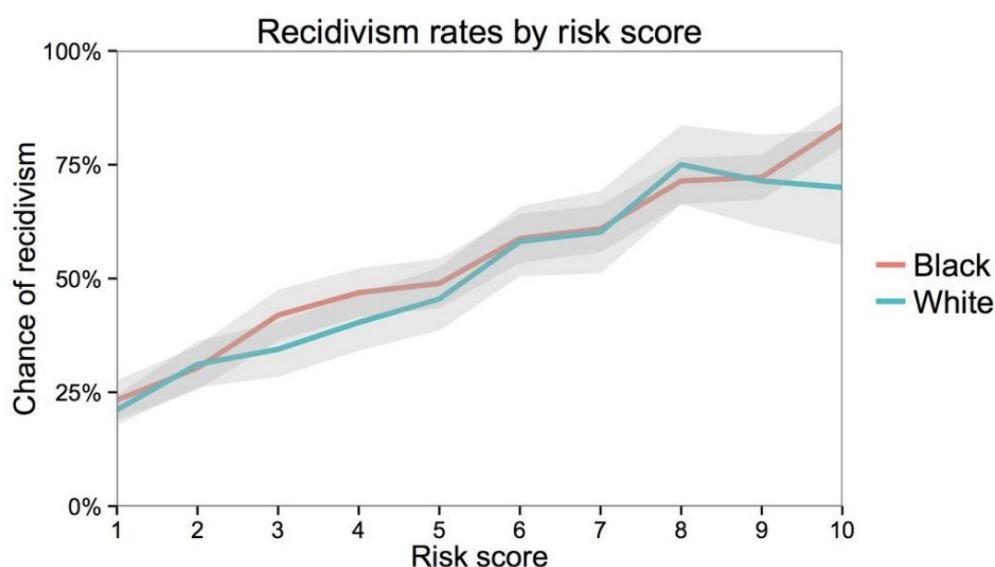
Judges have used COMPAS risk scores to help inform decisions about **bail, sentencing, and parole**.

WHAT HAPPENED?

In 2016, investigative journalists at **ProPublica** analyzed COMPAS risk scores for thousands of defendants and compared them to actual reoffense outcomes.

They found that the algorithm was significantly more likely to incorrectly flag Black defendants as **"high risk"** of reoffending, while white defendants who went on to reoffend were more often incorrectly labeled **"low risk."**

Both groups were roughly equally likely to actually reoffend, but the algorithm's errors were distributed unevenly across race.



WHY THIS WAS AN ETHICAL CONCERN

1. Bias

The training data reflected decades of policing and sentencing patterns that already contained racial disparities, so the model inherited and reinforced those patterns.

2. Transparency

COMPAS was a proprietary, commercially owned algorithm. Its exact scoring formula was not disclosed, so defendants and even judges could not fully understand or challenge how a score was calculated.

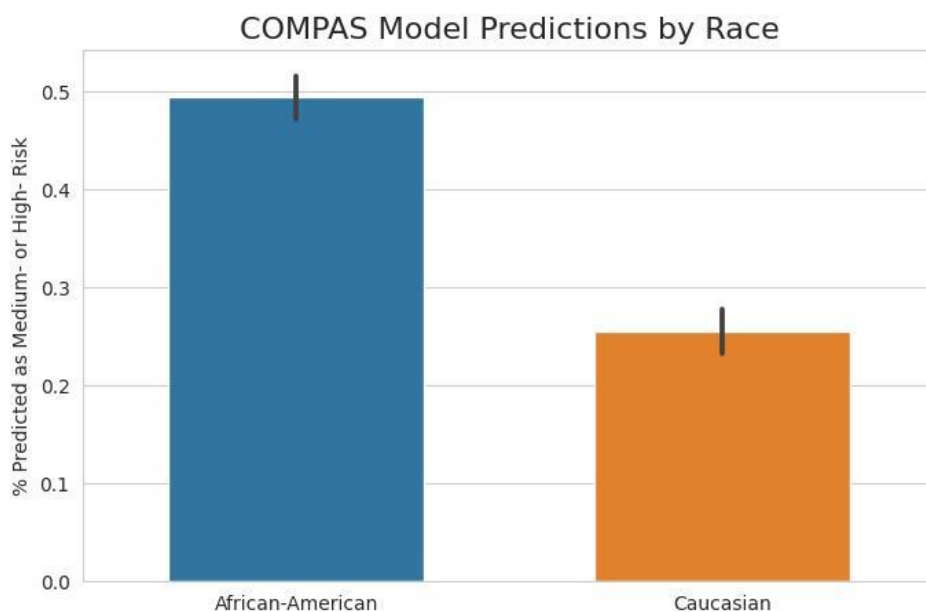
3. Accountability

The case raised the question of who is responsible when an opaque algorithm influences a person's freedom — the software company that built it, the court that used it, or neither.

Important Note

The methodology and conclusions of the ProPublica investigation were later debated by researchers and by the company behind COMPAS (Northpointe/Equivant), who argued the tool was "**fair**" under a different statistical definition of fairness.

This disagreement itself illustrates an important ethical lesson — **fairness can be defined in multiple, sometimes mathematically incompatible ways, and choosing which definition to prioritize is itself an ethical decision.**



TASK 3: MEASURES TO ADDRESS THESE ISSUES

A. ADDRESSING BIAS

- Audit training data for representation gaps and historical bias before development begins.
- Use fairness metrics during evaluation, such as **demographic parity** and **equalized odds**, instead of relying on accuracy alone.
- Involve diverse teams and representatives of affected communities in the design and review process.

B. ADDRESSING PRIVACY VIOLATIONS

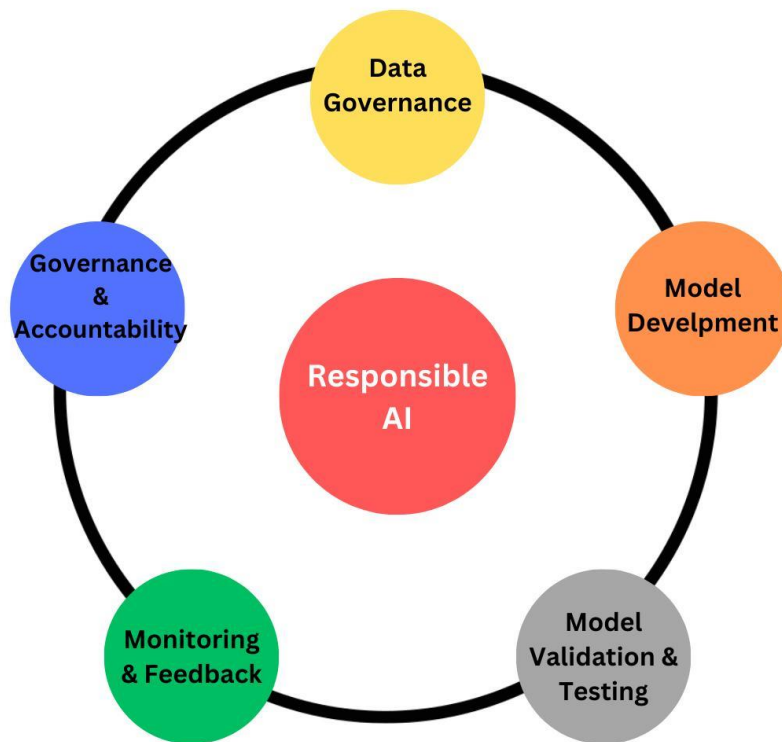
- Apply privacy-preserving techniques such as **differential privacy** and **federated learning** to reduce exposure of individual data.
- Obtain clear, informed consent and provide accessible opt-out mechanisms wherever possible.
- Follow the **data minimization principle** — collect and retain only the data that is strictly necessary.

C. ADDRESSING LACK OF TRANSPARENCY

- Use **Explainable AI (XAI)** techniques such as **SHAP** or **LIME** to make model decisions interpretable to humans.
- Require independent audits or regulatory oversight for AI systems used in high-stakes decisions.
- Publish documentation such as **model cards** or **datasheets for datasets** that disclose intended use, limitations, and known biases.

D. CROSS-CUTTING / GENERAL MEASURES

- Establish clear legal and organizational accountability frameworks for AI-driven decisions.
- Keep a **human reviewer in the loop** for high-impact decisions rather than fully automating them.
- Support regulation and industry standards, such as the **EU AI Act**, that mandate risk assessments for sensitive AI applications.



CONCLUSION

Deep Learning systems carry enormous potential, but the COMPAS case shows how quickly that potential can turn into real-world harm when bias, opacity, and weak accountability go unchecked.

Addressing these issues requires action at every stage — from how data is collected, to how models are tested, to how their decisions are explained and overseen.

Building trustworthy AI is not a one-time fix but an ongoing responsibility shared by **developers, organizations, and regulators.**